

LAB 2: AGILE BACKLOG CREATION & SPRINT SIMULATION

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PROJECT : MUNICIPAL SOLID WASTE AND ROUTE OPTIMISER

STORY POINTS & SPRINT EVIDENCE

Report: SCRUM Sprint 1 *Work item added after sprint start

Scope changes log View in All work navigator

Date ↑	Key	Summary	Work type	Epic	Details of scope change	Change in estimation
2026-08-28	SCRUM-8	Receive Bin Fill-Level Data	Story	Smart Bin M...	Estimate of 5 has been added	- → 5
2026-08-28	SCRUM-9	Identify Bins Requiring Collection	Story	Smart Bin M...	Estimate of 3 has been added	- → 3
2026-08-28	SCRUM-11	Generate Optimized Collection Route	Story	Route Optimiz...	Estimate of 8 has been added	- → 8
2026-08-28	SCRUM-13	Assign Route to Truck Driver	Story	Route Optimiz...	Estimate of 5 has been added	- → 5
2026-08-28	SCRUM-14	Update Collection Status	Story	Route Optimiz...	Estimate of 5 has been added	- → 5
2026-08-28	SCRUM-11*	Generate Optimized Collection Route	Story	Route Optimiz...	Work item added to sprint	8
2026-08-28	SCRUM-13*	Assign Route to Truck Driver	Story	Route Optimiz...	Work item added to sprint	5
2026-08-28	SCRUM-14*	Update Collection Status	Story	Route Optimiz...	Work item added to sprint	5
2026-08-28	SCRUM-8*	Receive Bin Fill-Level Data	Story	Smart Bin M...	Work item added to sprint	5
2026-08-28	SCRUM-9*	Identify Bins Requiring Collection	Story	Smart Bin M...	Work item added to sprint	3

Your sprint commitment has increased by 26 story points
Due to scope changes: You have 26 story points to complete this sprint

Completed work items View in All work navigator

Key	Summary	Work type	Epic	Status	Assignee	Story points
SCRUM-11	Generate Optimized Collection Route	Story	Route Optimiz...	Done		8
SCRUM-13	Assign Route to Truck Driver	Story	Route Optimiz...	Done		5
SCRUM-14	Update Collection Status	Story	Route Optimiz...	Done		5
SCRUM-8	Receive Bin Fill-Level Data	Story	Smart Bin Mo...	Done		5
SCRUM-9	Identify Bins Requiring Collection	Story	Smart Bin Mo...	Done		3

Spaces

Municipal Solid Waste Route Optimizer  ...

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☐ **SCRUM Sprint 1** 2 Sep – 9 Sep (1 work item)

0 0 3

 Complete sprint ...

Identify bins requiring waste collection and track their collection process.

☒ **SCRUM-17** Identify Bins Requiring Collection

Done ▾

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☐ **SCRUM Sprint 2** 2 Sep – 9 Sep (4 work items)

3 0 10

 Complete sprint ...

Improve waste collection route optimization and truck management by completing the planned user stories.

 SCRUM-18 Receive Bin Fill-Level Data	To Do ▾	3	= 
 SCRUM-21 Route Optimization & Truck Management	Done ▾	2	= 
 SCRUM-19 Generate Optimized Collection Routes	Done ▾	5	= 
 SCRUM-20 Assign Optimized Routes to Drivers	Done ▾	3	= 

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Sprint Execution & Simulation

Sprints were simulated in Jira to model daily progress across the active board:

- Sprint Board View: Tasks were systematically moved from To Do → In Progress → Done.

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work item details

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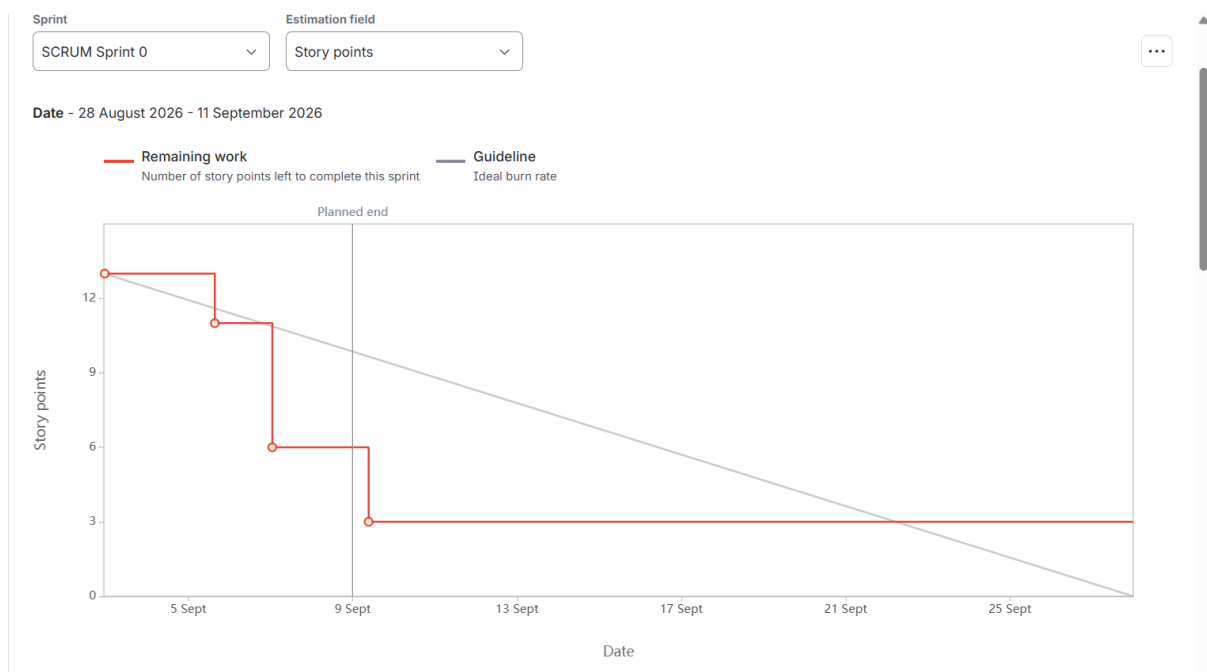
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Work item key	Work type	Work item priority	Issue summary
SCRUM-20	Story	Medium	Assign Optimized Routes to Drivers
SCRUM-21	Story	Medium	Route Optimization & Truck Management
SCRUM-4	Subtask		Subtask 2.1
SCRUM-5	Epic	Medium	Smart Bin Monitoring
SCRUM-6	Epic	Medium	Route Optimization & Truck Management
SCRUM-7	Epic	Medium	Missed Pickup Complaint Management
SCRUM-8	Story	High	Receive Bin Fill-Level Data
SCRUM-9	Story	High	Identify Bins Requiring Collection

Burndown Chart Analysis

The Jira burndown chart tracks total remaining work against an ideal linear burn rate for the selected sprint.

- **Daily Progression:** The chart shows remaining work decreasing from 13 to 11, then 6, then 3 story points, with step-downs on Sept 5, Sept 7, and Sept 9.
- **Capacity Insight:** The step-down pattern shows that work was completed incrementally across multiple days. The chart reaches 3 remaining story points by Sept 9, while the ideal guideline continues toward the planned sprint end.



Reflection Questions & Answers

Question 1: Did your estimations reflect the actual effort?

Answer:

Yes, using the Fibonacci sequence allowed us to estimate relative complexity rather than absolute time. Core operational logic like *Generate Optimized Collection Route* (8 story points) was estimated higher due to algorithm complexity and edge cases, whereas standard operational items like *Route Optimization & Truck Management* (2 story points) required significantly less effort.

Question 2: Was your backlog well-prioritized?

Answer:

Yes, the backlog was structured to deliver high-value functional dependencies first. Priority levels (High vs. Medium) ensured that foundational data collection features (e.g., smart bin fill-level ingestion) were completed before downstream route calculation and driver assignment features were executed.

Question 3: How did your simulated sprint align with your plan?

Answer:

The simulation closely followed our planned roadmap. Work items progressed standardly through the workflow columns (**To Do** → **In Progress** → **Done**) over consecutive simulated days. Out of the total committed story points, the high-priority core workflow items were successfully moved to **Done**, leaving minor non-blocking items in **To Do** for subsequent iterations.

Question 4: What insights did the burndown chart give about your team's capacity?

Answer:

The burndown chart shows steady progress, with remaining work decreasing from 13 to 11, 6, and 3 story points as user stories were completed. This indicates that work was completed incrementally across the simulated sprint.